North Hollywood High School

Dynamic Planet C

Key

CSE 2025

/90

Instructions:

- Do not open the test until time starts.
- You are permitted a binder and a calculator (without internet access).
- You are required to show all your work!
- All numerical answers should be reported to appropriate significant digits.
- Good luck!

Section 1: Plate Tectonics (10)

1. Which of the following features cannot form at a divergent plate boundary? (1)
    a. Mid-ocean ridge
    b. Volcanoes
    c. Rift valley
    d. Trench

2. Which of the following correctly matches the landform to its stage in the Wilson Cycle? (1)
    a. Red Sea, Embryonic
    b. Pacific Ocean, Declining
    c. Mediterranean, Suturing
    d. Atlantic, Declining

3. Which of the following is true about plate motion? (1)
    a. Slab suction is the driving force of motion
    b. Basal drag only resists plate motion
    c. Plates on average move around 1-2 meters per year
    d. Plate motion along mid ocean ridges are around equal on both sides

4. Explain what a fracture zone is and the difference between it and a transform fault. (3)

Inactive extensions of transform faults (1), transform are active boundaries, plates moving opposite directions (1)

5. How do marine magnetic anomalies on the seafloor prove the existence of seafloor spreading? (2)

Show symmetrical, alternating bands of magnetic polarity on both sides (1), prove new seafloor is created and spread outward (1)

Section 2: Marine Provinces (10)

1. Which of the following features indicate a passive margin? (1)
    a. Continental shelf
    b. Continental slope
    c. Continental rise
    d. Trench

2. What rock commonly makes up turbidite deposits? (1)
    a. Greywacke
    b. Breccia
    c. Limestone
    d. Sandstone

3. Which of the following cover the majority of Earth's surface? (1)
    a. Continental rises
    b. Abyssal plains
    c. Mid ocean ridges
    d. Continents

4. What types of sediments are most common from hydrothermal vents?
    a. Terrigenous
    b. Biogenous
    c. Hydrogenous
    d. Cosmogenous

5. Explain the difference and formation of seamounts and guyots. (4)

Seamounts cone shaped underwater volcanoes (1), guyots flat topped underwater volcanoes(1), guyots form from seamounts (1) that were once above sea level but tops were eroded (1)

6. Below is an image of the Hawaiian Islands. With the knowledge that the largest island is the youngest, explain the formation of these islands and how you can deduce the direction of movement of the tectonic plate. (3)



Formed over a hotspot (1), direction is measured youngest to oldest (1), so Northwest(1)

Section 3: Seawater (9)

1. Which of the following is closest to the salinity of seawater? (1)
   a. 0.035%
   b. 0.7%
   c. 3.5%
   d. 7%

2. Which of the following have the lowest concentration in seawater? (1)
   a. Potassium
   b. Sulfate
   c. Magnesium
   d. Nitrogen

3. Which of the following latitudes have the highest salinity? (1)
   a. 0
   b. 15 N
   c. 30 N
   d. 45 N

4. If I mix two parcels of water together with the same density but different temperatures and salinities, how will the density of the mixed parcel compare to the original parcels? Why is this? (3)

greater(1) cabbelling effect (1) nonlinear mixing of temperature and salinity lead to higher density(1)

5. Deep ocean water contains more CO_2 than surface water, but it is not highly acidic. Why does deep-ocean water contain more CO_2 and why is it not highly acidic? (3)

More Co2 because colder (1), colder means it holds more co2 so less becomes carbonic acid (1), also carbonate shells dissolves since under CCD, neutralizing some of the acid (1)

Section 4: Waves (10)

1. Which of the following do not contribute to wave speed? (1)
   a. Fetch distance
   b. Time of day
   c. Wind speed

d. Water depth

2. Beaufort Wind Scale describes sea surface conditions by which of the following? (1)

a. Wind speed
b. Wave frequency
c. Wave height
d. Wave speed

3. Which of the following are necessary for the formation of a rogue wave? (1)

a. Stormy weather
b. Earthquakes
c. High tides
d. Wave interference

4. There is a tsunami with a depth of 90 meters. What is the speed of this tsunami in meters per second? Is this a realistic speed for a tsunami in the ocean? What about near shore? (4)

sqrt(90*g) (1), 30m/s(1), not realistic for open ocean (1), realistic for near shore (1)

5. How are internal waves formed and where are they naturally found. (3)

Form where water densities meet, pycnocline (1), triggered by currents, tides or wind (1), found along continental slopes or other disruptive topographic features (1)

Section 5: Ocean Circulation (13)

1. About what percentage of wind energy is transferred to the ocean through friction? (1)
   a. 1%
   b. 2%
   c. 5%
   d. 10%

2. What is the largest gyre? (1)
   a. North Atlantic Subtropical Gyre
   b. North Pacific Subtropical Gyre
   c. South Atlantic Subtropical Gyre
   d. South Pacific Subtropical Gyre

3. Which of the following latitudes have the most pronounced upwelling? (1)
   a. 0
   b. 30N
   c. 60N
   d. 90N

4. Geostrophic Currents are balanced by gravity and what other force? (1)
   a. Pressure gradient force
   b. Coriolis Force
   c. Frictional force
   d. Wind stress

5. What are the major differences between western and eastern currents and what causes this difference? (3)

Western narrower, faster, warmer (1), eastern wider, slower, cooler (1), caused by earth's counterclockwise rotation compressing western currents

6. How does El Nino affect upwelling along the eastern pacific, equatorial currents, and the thermocline in the Pacific? (3)

Decreased upwelling in eastern pacific (1), reverse or weakens equatorial currents (1), deepens thermocline in eastern pacific, leveling the thermocline throughout the pacific (1)

7. What is Langmuir Circulation and how can it be observed at the surface? (3)

Shallow, circular surface currents caused by wind (1), windrows/streaks of foam or debris (1) parallel to wind (1)

Section 6: Tides (10)

1. To what power are tide-generating forces proportional to distance? (1)
   a. 0
   b. -1
   c. -2
   d. -3

2. How long is a tidal day? (1)
   a. 25 hours
   b. 24 hours 50 mins
   c. 24 hours 25 mins
   d. 23 hours 56 mins

3. How much of the Moon's tide-generating force does the Sun exert on Earth? (1)
   a. 25%
   b. 50%
   c. 100%
   d. 150%

4. Between about what latitudes do tidal bulges alternate? (1)
   a. 5N and 5S
   b. 10N and 10S
   c. 20N and 20S
   d. 30N and 30S

5. What is an amphidromic point and in what direction do tides rotate around these points? (3)

Points of zero tidal movement (1), tides rotate counterclockwise in northern hemisphere (1), clockwise in southern hemisphere (1)

6. What causes tidal inequality and how does it affect the tidal pattern at a location? (3)

Moon's tidal bulge shifting north and south of the equator (1), declination of moon and sun changing (1), sun's tide-generating forces interacting with moon, creating constructive or destructive interference (1)

Section 7: Coastal Processes (9)

1. Which of the following are found the closest to the lagoon on a barrier island? (1)
   a. Barrier flat
   b. Dune
   c. Salt marsh
   d. Beach

2. Which of the following are erosional features? (1)
   a. Sea caves
   b. Tombolo
   c. Spit
   d. Barrier island

3. What is the pattern of movement of sediment by longshore current? (1)
   a. Straight
   b. Zigzag
   c. Curved
   d. Circular

4. Which of the following stabilization techniques do not cause increased erosion elsewhere? (1)
   a. Beach replenishment
   b. Groins
   c. Breakwaters
   d. Jetties

5. How do the formation of glaciers affect eustatic and isostatic adjustment and how does this affect the overall sea level? (3)

Lowers eustatic sea levels (1), lowers isostatic levels (1), overall decreases sea level (1)

6. Where do waves normally refract toward and how does this affect coastlines over time? (2)

Refract toward headlands/protruding parts of the coast (1), leads to waves eroding these protrusions and creating a straight coastline (1)

Section 8: Marine Sediments (10)

1. What is the most common type of marine sediment? (1)
   - **a. Lithogenous**
   - b. Biogenous
   - c. Hydrogenous
   - d. Cosmogenous

2. If primary productivity along the equator increases, which type of sediment will show the greatest increase in concentration? (1)
   - a. Tektites
   - b. Evaporites
   - c. Calcareous Ooze
   - **d. Siliceous Ooze**

3. Which of the following are not found in pelagic deposits? (1)
   - a. Calcareous Ooze
   - b. Siliceous Ooze
   - **c. Stromatolites**
   - d. Manganese nodules

4. Which of the following types of deposition are most common for red clays? (1)
   - a. Turbidity currents
   - **b. Suspension settling**
   - c. Hydrothermal deposition
   - d. Ice-rafted debris

5. How does temperature affect the formation of calcareous ooze? And how does climate both impact the formation of calcareous ooze? (3)

warmer waters allow calcareous ooze to accumulate and vice versa (1), however climate change increases ocean acidity (1), leading to decreased formation of calcareous ooze (1)

6. How are manganese nodules formed on the seafloor? What marine structures are involved and commonly in what location? (3)

precipitation of metal oxides (1), around a nucleus (shell/rock) (1), grow commonly in the abyssal plains(1, 0.5 for seafloor)

Section 9: Climate Change (9)

1. Which of the following is not a major effect of climate change? (1)
   a. Increased coastal erosion
   b. Decreased Earth albedo
   c. Increased ocean salinity
   d. Decreased carbonate deposition

2. Which of the following is most responsible for sea level rise from global warming? (1)
   a. Thermal expansion of seawater
   b. Increased volcanic activity
   c. Decreased precipitation
   d. Increased river discharge

3. Which of the following is not a direct effect of climate change on ocean chemistry? (1)
   a. Ocean acidification
   b. Increased hypoxia in deep waters
   c. Decreased salinity due to melting ice
   d. Increased primary productivity

4. How do melting polar ice caps affect thermohaline circulation? (3)

Add freshwater to high latitudes (1) reduces salinity and density (1) slows down or disrupts thermohaline circulation (1)

5. How does climate change affect coral ecosystems? What symbiotic organism in coral is primarily affected and through what process? (3)

Climate change raises temperature (1), causing coral bleaching (1), zooxanthellae algae is expelled in the process (1)